then F maps the unit disc conformally onto the interior of a regular polygon with n sides and perimeter

$$2^{\frac{n-2}{n}} \int_0^{\pi} (\sin\theta)^{-2/n}\, d\theta.$$

24. The elliptic integrals K and K' defined for $0 < k < 1$ by

$$K(k) = \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \quad \text{and} \quad K'(k) = \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}}$$

satisfy various interesting identities. For instance:

(a) Show that if $\tilde{k}^2 = 1 - k^2$ and $0 < \tilde{k} < 1$, then

$$K'(k) = K(\tilde{k}).$$

[Hint: Change variables $x = (1 - \tilde{k}^2 y^2)^{-1/2}$ in the integral defining $K'(k)$.]

(b) Prove that if $\tilde{k}^2 = 1 - k^2$, and $0 < \tilde{k} < 1$, then

$$K(k) = \frac{2}{1+\tilde{k}}\, K\left(\frac{1-\tilde{k}}{1+\tilde{k}}\right).$$

[Hint: Change variables $x = 2t/(1 + \tilde{k} + (1-\tilde{k})t^2)$.]

(c) Show that for $0 < k < 1$ one has

$$K(k) = \frac{\pi}{2}\, F(1/2, 1/2, 1; k^2),$$

where F the hypergeometric series. [Hint: This follows from the integral representation for F given in Exercise 9, Chapter 6.]

6 Problems

1. Let f be a complex-valued C^1 function defined in the neighborhood of a point z_0. There are several notions closely related to conformality at z_0. We say that f is **isogonal** at z_0 if whenever $\gamma(t)$ and $\eta(t)$ are two smooth curves with $\gamma(0) = \eta(0) = z_0$, that make an angle θ there ($|\theta| < \pi$), then $f(\gamma(t))$ and $f(\eta(t))$ make an angle of θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ. Also, f is said to be **isotropic** if it magnifies lengths by some factor for all directions emanating from z_0, that is, if the limit

$$\lim_{r\to 0} \frac{|f(z_0 + re^{i\theta}) - f(z_0)|}{r}$$

exists, is non-zero, and independent of θ.

Then f is isogonal at z_0 if and only if it is isotropic at z_0; moreover, f is isogonal at z_0 if and only if either $f'(z_0)$ exists and is non-zero, or the same holds for f replaced by $\overline{f}$.

2. The angle between two non-zero complex numbers z and w (taken in that order) is simply the oriented angle, in $(-\pi, \pi]$, that is formed between the two vectors in $\mathbb{R}^2$ corresponding to the points z and w. This oriented angle, say α, is uniquely determined by the two quantities

$$\frac{(z, w)}{|z|\,|w|} \quad \text{and} \quad \frac{(z, -iw)}{|z|\,|w|}$$

which are simply the cosine and sine of α, respectively. Here, the notation $(\cdot, \cdot)$ corresponds to the usual Euclidian inner product in $\mathbb{R}^2$, which in terms of complex numbers takes the form $(z, w) = \mathrm{Re}(z\overline{w})$.

In particular, we may now consider two smooth curves $\gamma : [a, b] \to \mathbb{C}$ and $\eta : [a, b] \to \mathbb{C}$, that intersect at z_0, say $\gamma(t_0) = \eta(t_0) = z_0$, for some $t_0 \in (a, b)$. If the quantities $\gamma'(t_0)$ and $\eta'(t_0)$ are non-zero, then they represent the tangents to the curves γ and η at the point z_0, and we say that the two curves intersect at z_0 at the angle formed by the two vectors $\gamma'(t_0)$ and $\eta'(t_0)$.

A holomorphic function f defined near z_0 is said to **preserve angles** at z_0 if for any two smooth curves γ and η intersecting at z_0, the angle formed between the curves γ and η at z_0 equals the angle formed between the curves $f \circ \gamma$ and $f \circ \eta$ at $f(z_0)$. (See Figure 12 for an illustration.) In particular, we assume that the tangents to the curves γ, η, $f \circ \gamma$, and $f \circ \eta$ at the point z_0 and $f(z_0)$ are all non-zero.

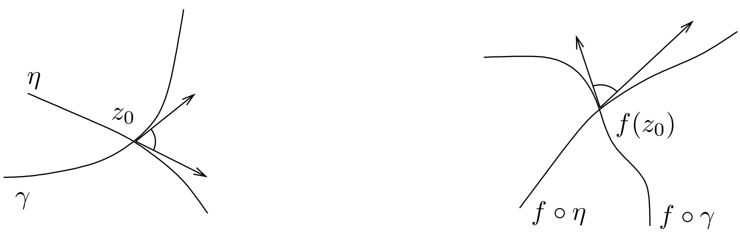

Figure 12. Preservation of angles at z_0

(a) Prove that if $f : \Omega \to \mathbb{C}$ is holomorphic, and $f'(z_0) \neq 0$, then f preserves angles at z_0. [Hint: Observe that

$$(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0), \eta'(t_0)).]$$

(b) Conversely, prove the following: suppose $f : \Omega \to \mathbb{C}$ is a complex-valued function, that is real-differentiable at $z_0 \in \Omega$, and $J_f(z_0) \neq 0$. If f preserves angles at z_0, then f is holomorphic at z_0 with $f'(z_0) \neq 0$.

3.* The Schwarz-Pick lemma (see Exercise 13) is the infinitesimal version of an important observation in complex analysis and geometry.

For complex numbers $w \in \mathbb{C}$ and $z \in \mathbb{D}$ we define the **hyperbolic length** of w at z by

$$\|w\|_z = \frac{|w|}{1 - |z|^2},$$

where $|w|$ and $|z|$ denote the usual absolute values. This length is sometimes referred to as the **Poincaré metric**, and as a Riemann metric it is written as

$$ds^2 = \frac{|dz|^2}{(1 - |z|^2)^2}.$$

The idea is to think of w as a vector lying in the tangent space at z. Observe that for a fixed w, its hyperbolic length grows to infinity as z approaches the boundary of the disc. We pass from the infinitesimal hyperbolic length of tangent vectors to the global hyperbolic distance between two points by integration.

(a) Given two complex numbers z_1 and z_2 in the disc, we define the **hyperbolic distance** between them by

$$d(z_1, z_2) = \inf_{\gamma} \int_0^1 \|\gamma'(t)\|_{\gamma(t)}\, dt,$$

where the infimum is taken over all smooth curves $\gamma : [0, 1] \to \mathbb{D}$ joining z_1 and z_2. Use the Schwarz-Pick lemma to prove that if $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then

$$d(f(z_1), f(z_2)) \leq d(z_1, z_2) \qquad \text{for any } z_1, z_2 \in \mathbb{D}.$$

In other words, holomorphic functions are distance-decreasing in the hyperbolic metric.

(b) Prove that automorphisms of the unit disc preserve the hyperbolic distance, namely

$$d(\varphi(z_1), \varphi(z_2)) = d(z_1, z_2), \qquad \text{for any } z_1, z_2 \in \mathbb{D}$$

and any automorphism φ. Conversely, if $\varphi : \mathbb{D} \to \mathbb{D}$ preserves the hyperbolic distance, then either φ or $\overline{\varphi}$ is an automorphism of $\mathbb{D}$.

(c) Given two points $z_1, z_2 \in \mathbb{D}$, show that there exists an automorphism φ such that $\varphi(z_1) = 0$ and $\varphi(z_2) = s$ for some s on the segment $[0, 1)$ on the real line.

(d) Prove that the hyperbolic distance between 0 and $s \in [0, 1)$ is

$$d(0, s) = \frac{1}{2} \log \frac{1 + s}{1 - s}.$$

(e) Find a formula for the hyperbolic distance between any two points in the unit disc.

4.* Consider the group of matrices of the form

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that satisfy the following conditions:

(i) a, b, c, and $d \in \mathbb{C}$,

(ii) the determinant of M is equal to 1,

(iii) the matrix M preserves the following hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$:

$$\langle Z, W \rangle = z_1 \overline{w}_1 - z_2 \overline{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. In other words, for all $Z, W \in \mathbb{C}^2$

$$\langle MZ, MW \rangle = \langle Z, W \rangle.$$

This group of matrices is denoted by $\mathrm{SU}(1,1)$.

(a) Prove that all matrices in $\mathrm{SU}(1,1)$ are of the form

$$\begin{pmatrix} a & b \\ \overline{b} & \overline{a} \end{pmatrix},$$

where $|a|^2 - |b|^2 = 1$. To do so, consider the matrix

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and observe that $\langle Z, W \rangle = {}^t\overline{W} J Z$, where ${}^t\overline{W}$ denotes the conjugate transpose of W.

(b) To every matrix in $\mathrm{SU}(1,1)$ we can associate a fractional linear transformation

$$\frac{az+b}{cz+d}.$$

Prove that the group $\mathrm{SU}(1,1)/\{\pm 1\}$ is isomorphic to the group of automorphisms of the disc. [Hint: Use the following association.]

$$e^{2i\theta} \frac{z-\alpha}{1-\overline{\alpha}z} \longrightarrow \begin{pmatrix} \frac{e^{i\theta}}{\sqrt{1-|\alpha|^2}} & -\frac{\alpha e^{i\theta}}{\sqrt{1-|\alpha|^2}} \\ -\frac{\overline{\alpha} e^{-i\theta}}{\sqrt{1-|\alpha|^2}} & \frac{e^{-i\theta}}{\sqrt{1-|\alpha|^2}} \end{pmatrix}.$$

5. The following result is relevant to Problem 4 in Chapter 10 which treats modular functions.

(a) Suppose that $F : \mathbb{H} \to \mathbb{C}$ is holomorphic and bounded. Also, suppose that $F(z)$ vanishes when $z = ir_n$, $n = 1, 2, 3, \ldots$, where $\{r_n\}$ is a bounded sequence of positive numbers. Prove that if $\sum_{n=1}^{\infty} r_n = \infty$, then $F = 0$.

(b) If $\sum r_n < \infty$, it is possible to construct a bounded function on the upper half-plane with zeros precisely at the points ir_n.

For related results in the unit disc, see Problems 1 and 2 in Chapter 5.]

6.* The results of Exercise 18 extend to the case when γ is assumed merely to be closed, simple, and continuous. The proof, however, requires further ideas.

7.* Applying ideas of Carathéodory, Koebe gave a proof of the Riemann mapping theorem by constructing (more explicitly) a sequence of functions that converges to the desired conformal map.

Starting with a Koebe domain, that is, a simply connected domain $\mathcal{K}_0 \subset \mathbb{D}$ that is not all of $\mathbb{D}$, and which contains the origin, the strategy is to find an injective function f_0 such that $f_0(\mathcal{K}_0) = \mathcal{K}_1$ is a Koebe domain "larger" than $\mathcal{K}_0$. Then, one iterates this process, finally obtaining functions $F_n = f_n \circ \cdots \circ f_0 : \mathcal{K}_0 \to \mathbb{D}$ such that $F_n(\mathcal{K}_0) = \mathcal{K}_{n+1}$ and $\lim F_n = F$ is a conformal map from $\mathcal{K}_0$ to $\mathbb{D}$.

The **inner radius** of a region $\mathcal{K} \subset \mathbb{D}$ that contains the origin is defined by $r_{\mathcal{K}} = \sup\{\rho \geq 0 : D(0,\rho) \subset \mathcal{K}\}$. Also, a holomorphic injection $f : \mathcal{K} \to \mathbb{D}$ is said to be an **expansion** if $f(0) = 0$ and $|f(z)| > |z|$ for all $z \in \mathcal{K} - \{0\}$.

(a) Prove that if f is an expansion, then $r_{f(\mathcal{K})} \geq r_{\mathcal{K}}$ and $|f'(0)| > 1$. [Hint: Write $f(z) = zg(z)$ and use the maximum principle to prove that $|f'(0)| = |g(0)| > 1$.]

Suppose we begin with a Koebe domain $\mathcal{K}_0$ and a sequence of expansions $\{f_0, f_1, \ldots, f_n, \ldots\}$, so that $\mathcal{K}_{n+1} = f_n(\mathcal{K}_n)$ are also Koebe domains. We then define holomorphic maps $F_n : \mathcal{K}_0 \to \mathbb{D}$ by $F_n = f_n \circ \cdots \circ f_0$.

(b) Prove that for each n, the function F_n is an expansion. Moreover, $F_n'(0) = \prod_{k=0}^{n} f_k'(0)$, and conclude that $\lim_{n\to\infty} |f_n'(0)| = 1$. [Hint: Prove that the sequence $\{|F_n'(0)|\}$ has a limit by showing that it is bounded above and monotone increasing. Use the Schwarz lemma.]

(c) Show that if the sequence is osculating, that is, $r_{\mathcal{K}_n} \to 1$ as $n \to \infty$, then $\{F_n\}$ converges uniformly on compact subsets of $\mathcal{K}_0$ to a conformal map $F : \mathcal{K}_0 \to \mathbb{D}$. [Hint: If $r_{F(\mathcal{K}_0)} \geq 1$ then F is surjective.]

To construct the desired osculating sequence we shall use the automorphisms $\psi_\alpha = (\alpha - z)/(1 - \overline{\alpha} z)$.

(d) Given a Koebe domain $\mathcal{K}$, choose a point $\alpha \in \mathbb{D}$ on the boundary of $\mathcal{K}$ such that $|\alpha| = r_{\mathcal{K}}$, and also choose $\beta \in \mathbb{D}$ such that $\beta^2 = \alpha$. Let S denote the square root of ψ_α on $\mathcal{K}$ such that $S(0) = 0$. Why is such a function well defined? Prove that the function $f : \mathcal{K} \to \mathbb{D}$ defined by $f(z) = \psi_\beta \circ S \circ \psi_\alpha$

is an expansion. Moreover, show that $|f'(0)| = (1 + r_{\mathcal{K}})/2\sqrt{r_{\mathcal{K}}}$. [Hint: To prove that $|f(z)| > |z|$ on $\mathcal{K} - \{0\}$ apply the Schwarz lemma to the inverse function, namely $\psi_\alpha \circ g \circ \psi_\beta$ where $g(z) = z^2$.]

(e) Use part (d) to construct the desired sequence.

8.* Let f be an injective holomorphic function in the unit disc, with $f(0) = 0$ and $f'(0) = 1$. If we write $f(z) = z + a_2 z^2 + a_3 z^3 \cdots$, then Problem 1 in Chapter 3 shows that $|a_2| \leq 2$. Bieberbach conjectured that in fact $|a_n| \leq n$ for all $n \geq 2$; this was proved by deBranges. This problem outlines an argument to prove the conjecture under the additional assumption that the coefficients a_n are real.

(a) Let $z = re^{i\theta}$ with $0 < r < 1$, and show that if $v(r, \theta)$ denotes the imaginary part of $f(re^{i\theta})$, then

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin n\theta \, d\theta.$$

(b) Show that for $0 \leq \theta \leq \pi$ and $n = 1, 2, \ldots$ we have $|\sin n\theta| \leq n \sin \theta$.

(c) Use the fact that $a_n \in \mathbb{R}$ to show that $f(\mathbb{D})$ is symmetric with respect to the real axis, and use this fact to show that f maps the upper half-disc into either the upper or lower part of $f(\mathbb{D})$.

(d) Show that for r small,

$$v(r, \theta) = r \sin \theta [1 + O(r)],$$

and use the previous part to conclude that $v(r, \theta) \sin \theta \geq 0$ for all $0 < r < 1$ and $0 \leq \theta \leq \pi$.

(e) Prove that $|a_n r^n| \leq nr$, and let $r \to 1$ to conclude that $|a_n| \leq n$.

(f) Check that the function $f(z) = z/(1 - z)^2$ satisfies all the hypotheses and that $|a_n| = n$ for all n.

9.* Gauss found a connection between elliptic integrals and the familiar operations of forming arithmetic and geometric means.

We start with any pair (a, b) of numbers that satisfy $a \geq b > 0$, and form the arithmetic and geometric means of a and b, that is,

$$a_1 = \frac{a + b}{2} \qquad \text{and} \qquad b_1 = (ab)^{1/2}.$$

We then repeat these operations with a and b replaced by a_1 and b_1. Iterating this process provides two sequences $\{a_n\}$ and $\{b_n\}$ where a_{n+1} and b_{n+1} are the arithmetic and geometric means of a_n and b_n, respectively.

(a) Prove that the two sequences $\{a_n\}$ and $\{b_n\}$ have a common limit. This limit, which we denote by $M(a, b)$, is called the **arithmetic-geometric mean** of a and b. [Hint: Show that $a \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq b_n \geq \cdots \geq b_1 \geq b$ and $a_n - b_n \leq (a - b)/2^n$.]

(b) Gauss's identity states that

$$\frac{1}{M(a,b)} = \frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)^{1/2}}.$$

To prove this relation, show that if $I(a, b)$ denotes the integral on the right-hand side, then it suffices to establish the invariance of I, namely

$$I(a,b) = I\left(\frac{a+b}{2}, (ab)^{1/2}\right). \tag{6}$$

Then, observe that the connection with elliptic integrals takes the form

$$I(a,b) = \frac{1}{a} K(k) = \frac{1}{a} \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \qquad \text{where } k^2 = 1 - b^2/a^2,$$

and that the relation (6) is a consequence of the identity in (b) of Exercise 24.

9 An Introduction to Elliptic Functions

> The form that Jacobi had given to the theory of elliptic functions was far from perfection; its flaws are obvious. At the base we find three fundamental functions sn, cn and dn. These functions do not have the same periods...
>
> In Weierstrass' system, instead of three fundamental functions, there is only one, $\wp(u)$, and it is the simplest of all having the same periods. It has only one double infinity; and finally its definition is so that it does not change when one replaces one system of periods by another equivalent system.
>
> *H. Poincaré,* 1899

The theory of elliptic functions, which is of interest in several parts of mathematics, initially grew out of the study of elliptic integrals. These can be described generally as integrals of the form $\int R(x, \sqrt{P(x)})\, dx$, where R is a rational function and P a polynomial of degree three or four.[1] These integrals arose in computing the arc-length of an ellipse, or of a lemniscate, and in a variety of other problems. Their early study was centered on their special transformation properties and on the discovery of an inherent double-periodicity. We have seen an example of this latter phenomenon in the mapping function of the half-plane to a rectangle taken up in Section 4.5 of the previous chapter.

It was Jacobi who transformed the subject by initiating the systematic study of doubly-periodic functions (called elliptic functions). In this theory, the theta functions he introduced played a decisive role. Weierstrass after him developed another approach, which in its initial steps is simpler and more elegant. It is based on his $\wp$ function, and in this chapter we shall sketch the beginnings of that theory. We will go as far as to glimpse a possible connection with number theory, by considering the Eisenstein series and their expression involving divisor functions. A number of more direct links with combinatorics and number theory arise from the theta

[1] The case when P is a quadratic polynomial is essentially that of "circular functions", and can be reduced to the trigonometric functions $\sin x$, $\cos x$, etc.